

2201NSC Linear Algebra and Applications

Week 1

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What is Linear Algebra?

A **linear** operation $\mathcal{L}$ is one that obeys the relationship

$$\mathcal{L}(\alpha x + \beta y) = \alpha \mathcal{L}(x) + \beta \mathcal{L}(y)$$

where α, β are scalars.

Example of a Linear Operator: Differentiation

Let $\mathcal{D}$ be the differentiation operator:

$$\mathcal{D}(f(x)) = \frac{d}{dx}f(x)$$

Differentiation is a **linear operator** because it preserves linear combinations:

$$\mathcal{D}(\alpha f(x) + \beta g(x)) = \alpha \mathcal{D}(f(x)) + \beta \mathcal{D}(g(x))$$

For example,

$$\frac{d}{dx}(\alpha f(x) + \beta g(x)) = \alpha \frac{d}{dx}f(x) + \beta \frac{d}{dx}g(x).$$

Example of a Linear Operator: Integration

Let $\mathcal{I}$ be the integration operator:

$$\mathcal{I}(f(x)) = \int f(x) dx$$

Integration is a **linear operator** because it preserves linear combinations:

$$\mathcal{I}(\alpha f(x) + \beta g(x)) = \alpha \mathcal{I}(f(x)) + \beta \mathcal{I}(g(x))$$

For example,

$$\int (\alpha f(x) + \beta g(x)) dx = \alpha \int f(x) dx + \beta \int g(x) dx.$$

Simultaneous Equations

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Simultaneous Equations



Simultaneous Equations

System

$$\begin{aligned}x_1 + x_3 &= 4, \\ 2x_1 + x_2 - x_3 &= 1, \\ x_2 + x_3 &= 5.\end{aligned}$$

Augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 0 & 1 & 1 & 5 \end{array} \right)$$

- ▶ Fix the variable order before writing any row.
- ▶ Insert a zero when a variable is missing.

Three Row Operations

1. **Swap rows:** $R_i \leftrightarrow R_j$
2. **Scale a row:** $R_i \leftarrow cR_i$, where $c \neq 0$
3. **Replace a row:** $R_i \leftarrow R_i + cR_j$

Different valid row-operation paths lead to the same solution.

Worked example: eliminate first, then back-substitute

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 0 & 1 & 1 & 5 \end{array} \right) \xrightarrow[R_3 \text{ unchanged}]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & -3 & -7 \\ 0 & 1 & 1 & 5 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & -3 & -7 \\ 0 & 0 & 4 & 12 \end{array} \right).$$

$$\begin{aligned} 4x_3 &= 12 & \Rightarrow x_3 &= 3, \\ x_2 - 3x_3 &= -7 & \Rightarrow x_2 &= 2, \\ x_1 + x_3 &= 4 & \Rightarrow x_1 &= 1. \end{aligned}$$

Unique solution

$$(x_1, x_2, x_3) = (1, 2, 3).$$

Row echelon form

A matrix is in **row echelon form** when:

1. all zero rows are at the bottom;
2. each leading entry is to the right of the one above;
3. all entries below each leading entry are zero.

$$\left(\begin{array}{cccc|c} * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Leading entries identify **lead variables**; non-pivot columns identify **free variables**.

Determine the type of solution

Unique Solution

No contradiction, and every variable column has a pivot.

$$\left(\begin{array}{cc|c} * & * & * \\ 0 & * & * \end{array} \right)$$

No Solution

A row demands an impossibility.

$$\left(\begin{array}{cc|c} * & * & * \\ 0 & 0 & c \end{array} \right), \quad c \neq 0$$

Family of Solutions

At least one variable is free.

$$\left(\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \end{array} \right)$$

Worked example

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 3 & 3 \end{array} \right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \textcolor{red}{1} \end{array} \right).$$

Read the last row as an equation

$$0x_1 + 0x_2 = 1 \quad \Longleftrightarrow \quad 0 = 1.$$

The system is **inconsistent** and has **no solution**.

Worked example

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{array} \right).$$

Pivot columns: x_1, x_2

Free column: x_3

Family of solutions

$$x_1 = 1, \quad x_2 + x_3 = 0.$$

$$(x_1, x_2, x_3) = (1, -t, t), \quad t \in \mathbb{R}.$$

Set $x_3 = t$, where $t \in \mathbb{R}$.

REF vs RREF (Reduced Row Echelon Form)

Gaussian elimination

Stop at **row echelon form**.

- ▶ Efficient by hand
- ▶ Requires back-substitution

Gauss–Jordan elimination

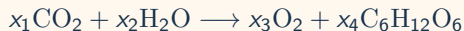
Continue to **reduced row echelon form**.

- ▶ Each pivot is 1
- ▶ A pivot is the only nonzero entry in its column
- ▶ Solutions can be read directly

Software check

In MATLAB/Octave use `rref([A b])`

Application: balancing photosynthesis gives a free scale



Conservation of C, H and O gives

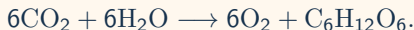
$$x_1 - 6x_4 = 0,$$

$$2x_2 - 12x_4 = 0,$$

$$2x_1 + x_2 - 2x_3 - 6x_4 = 0.$$

Taking $x_4 = t$ gives $x_1 = 6t$, $x_2 = 6t$, $x_3 = 6t$. Choose the smallest positive integer scale, $t = 1$:

Balanced equation



Question 1

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 2 & 6 \end{array} \right)$$

REF

RREF

Neither

Question 2

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & 5 \end{array} \right)$$

REF

RREF

Neither

Question 3

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right)$$

REF

RREF

Neither

Question 4

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 1 & 2 & 5 \end{array} \right)$$

REF

RREF

Neither

Question 5

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 & 2 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

REF

RREF

Neither

Question 6

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

REF

RREF

Neither

Question 7

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 4 \\ 0 & 1 & 0 & -1 & 3 \\ 0 & 0 & 1 & 5 & 2 \\ 0 & 0 & 0 & 1 & 6 \end{array} \right)$$

REF

RREF

Neither

Question 8

$$\left(\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right)$$

REF

RREF

Neither

Question 9

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

REF

RREF

Neither

Answers

Q	Answer	Reason
1	REF	Pivots move right and entries below pivots are zero.
2	REF	The third pivot has nonzero entries above it, so it is not RREF.
3	RREF	Every pivot is 1 and is the only nonzero entry in its column.
4	Neither	The leading entry in row 3 is left of the leading entry in row 2.
5	REF	Pivots move right and zero rows are at the bottom.
6	RREF	Pivot columns are reduced and the zero row is at the bottom.
7	REF	The fourth pivot has nonzero entries above it.
8	Neither	The leading entries do not move consistently to the right.
9	REF	The row $[0 \ 0 \ 0 \mid 1]$ gives a contradiction, but the pivot column is not reduced.